

Thought for a couple of seconds >

We use the convention

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk.$$

1. Exponential decay

Given

$$f(x) = e^{-a|x|}, \quad a > 0,$$

split the integral at $x = 0$:

$$\tilde{f}(k) = \int_{-\infty}^0 e^{ax} e^{-ikx} dx + \int_0^{\infty} e^{-ax} e^{-ikx} dx.$$

The first integral is

$$\int_{-\infty}^0 e^{(a-ik)x} dx = \frac{1}{a-ik},$$

and the second is

$$\int_0^{\infty} e^{-(a+ik)x} dx = \frac{1}{a+ik}.$$

Therefore,

$$\tilde{f}(k) = \frac{1}{a-ik} + \frac{1}{a+ik} = \boxed{\frac{2a}{a^2 + k^2}}.$$

2. Gaussian

Given

$$f(x) = e^{-ax^2}, \quad a > 0,$$

the Fourier transform is

$$\tilde{f}(k) = \int_{-\infty}^{\infty} e^{-ax^2 - ikx} dx.$$

Complete the square:

$$-ax^2 - ikx = -a \left(x + \frac{ik}{2a} \right)^2 - \frac{k^2}{4a}.$$

Thus,

$$\tilde{f}(k) = e^{-k^2/(4a)} \int_{-\infty}^{\infty} e^{-a(x + \frac{ik}{2a})^2} dx.$$

The shifted Gaussian integral has the same value as the ordinary Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-a(x + \frac{ik}{2a})^2} dx = \sqrt{\frac{\pi}{a}}.$$

Therefore,

$$\boxed{\tilde{f}(k) = \sqrt{\frac{\pi}{a}} e^{-k^2/(4a)}}.$$

3. Rectangular pulse

Given

$$f(x) = \begin{cases} 1, & |x| \leq L, \\ 0, & |x| > L, \end{cases}$$

the Fourier transform is

$$\tilde{f}(k) = \int_{-L}^L e^{-ikx} dx.$$

Integrating,

$$\tilde{f}(k) = \left[\frac{e^{-ikx}}{-ik} \right]_{-L}^L = \frac{e^{-ikL} - e^{ikL}}{-ik}.$$

Using

$$e^{-ikL} - e^{ikL} = -2i \sin(kL),$$

we obtain

$$\boxed{\tilde{f}(k) = \frac{2 \sin(kL)}{k}}.$$

At $k = 0$, use

$$\lim_{k \rightarrow 0} \frac{\sin(kL)}{k} = L.$$

Therefore,

$$\boxed{\tilde{f}(0) = 2L}.$$

This also follows directly from

$$\tilde{f}(0) = \int_{-L}^L 1 \, dx = 2L.$$

4. Shifted function

Suppose

$$g(x) = f(x - x_0).$$

Its Fourier transform is

$$\tilde{g}(k) = \int_{-\infty}^{\infty} f(x - x_0) e^{-ikx} \, dx.$$

Introduce the variable

$$u = x - x_0, \quad x = u + x_0, \quad dx = du.$$

Then

$$\tilde{g}(k) = \int_{-\infty}^{\infty} f(u) e^{-ik(u+x_0)} \, du.$$

Factor out the part independent of u :

$$\tilde{g}(k) = e^{-ikx_0} \int_{-\infty}^{\infty} f(u) e^{-iku} \, du.$$

Therefore,

$$\boxed{\tilde{g}(k) = e^{-ikx_0} \tilde{f}(k)}.$$

For the shifted rectangular pulse,

$$g(x) = \begin{cases} 1, & |x - x_0| \leq L, \\ 0, & |x - x_0| > L, \end{cases}$$

we use the result of Problem 3:

$$\tilde{f}(k) = \frac{2 \sin(kL)}{k}.$$

Hence,

$$\tilde{g}(k) = e^{-ikx_0} \frac{2 \sin(kL)}{k}.$$

The factor e^{-ikx_0} changes only the phase because

$$|e^{-ikx_0}| = 1.$$

Thus,

$$|\tilde{g}(k)| = |\tilde{f}(k)|.$$

5. Derivative property

We want to evaluate

$$\mathcal{F} \left[\frac{df}{dx} \right] (k) = \int_{-\infty}^{\infty} \frac{df}{dx} e^{-ikx} dx.$$

Integrating by parts, take

$$u = e^{-ikx}, \quad dv = \frac{df}{dx} dx.$$

Then

$$du = -ike^{-ikx} dx, \quad v = f(x).$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{df}{dx} e^{-ikx} dx = [f(x)e^{-ikx}]_{-\infty}^{\infty} + ik \int_{-\infty}^{\infty} f(x)e^{-ikx} dx.$$

Since $f(x) \rightarrow 0$ as $x \rightarrow \pm\infty$, the boundary term vanishes. Hence,

$$\mathcal{F} \left[\frac{df}{dx} \right] (k) = ik \tilde{f}(k).$$

Now consider

$$h(x) = \frac{d}{dx} e^{-ax^2}.$$

Since

$$\frac{d}{dx}e^{-ax^2} = -2ax e^{-ax^2},$$

we can apply the derivative property directly. From Problem 2,

$$\mathcal{F}[e^{-ax^2}] = \sqrt{\frac{\pi}{a}}e^{-k^2/(4a)}.$$

Therefore,

$$\tilde{h}(k) = ik\sqrt{\frac{\pi}{a}}e^{-k^2/(4a)}.$$

Thus,

$$\boxed{\mathcal{F}\left[\frac{d}{dx}e^{-ax^2}\right] = ik\sqrt{\frac{\pi}{a}}e^{-k^2/(4a)}}.$$